Sample paper Physics (PHY101)

Exam Type: - Final/End Term Paper Course: - Physics – PHY101

Total Marks: - -60

Marks for Each Question	Total Questions in Paper	No. of Questions (to be Graded)	Choice Questions	Marks for Graded Questions(in each category)	Time Allocated Per Question (minutes)	Total Time Required for Graded Questions (minutes)
3	6	4	2	12	6	24
5	4	3	1	15	10	30
1	40	33	7	33	1	33
Total	50		10	60		87
Final Exam Session Duration						90

Dear students,

Note: The syllabus of Final/End term is video lecture # 19 to 45.

Here are some questions for sample.

Attempt any 33 MCQs from the following (some sample questions are given below)

MCQ # 01. **Suppose A = BC, where A has the dimension L/M and C has the dimension L/T. Then B has the dimension:**

- **a.** T/M
- **b.** L2/TM
- **c.** TM/L2
- **d.** L2T/M

MCQ # 02. **A bullet shoot straight up returns to its starting point in 10 sec. the initial speed was:**

- **a.** 9.8 m/s
- **b.** 24.5 m/s
- **c.** 49 m/s
- **d.** 98 m/s

MCQ # 03. **The acceleration of projectile in x-direction:**

- **a.** Depends on initial velocity
- **b.** Depends on how long it is in the air
- **c.** 0 m/s2
- **d.** Depends on y-acceleration

MCQ # 04. **The scalar product of two vectors is maximum when they are:**

- **a.** Parallel
- **b.** Perpendicular
- **c.** Null
- **d.** Anti-parallel

MCQ # 05. **The inertia of a body tends to cause the body to:**

a. speed up
b. slow down
c. resist any change in its motion
d. fall toward Earth

MCQ # 06. A force of 120 N is exerted on a 40 kg container which sits on a floor. If the frictional force between floor and container is 80 N, what is the magnitude of the acceleration of the container?

a. 3 ms-2
b. 2 ms-2
c. 1 ms-2
d. 5 ms-2

MCQ # 07. According to the equation of work, when a boy sits in one place and studies for the whole night, he does:

a. lot of work
b. no work
c. too much work
d. very little work

MCQ # 08. The work done by gravity during the descent of a projectile:

a. depends for its sign on the direction of the y axis
b. is positive
c. is negative
d. is zero

MCQ # 09. A 6.0-kg block is released from rest 80m above the ground. When it has fallen 60m its kinetic energy is approximately:

a. 4800 J
b. 3500 J
c. 1200 J
d. 120 J

MCQ # 10. Two bodies, A and B, have equal kinetic energies. The mass of A is nine times that of B. The ratio of the momentum of A to that of B is:

a. 1:9
b. 1:3
c. 1:1
d. 3:1

MCQ # 11. A rifle of mass M is initially at rest but free to recoil. It fires a bullet of mass m and velocity v (relative to the ground). After firing, the velocity of the rifle (relative to the ground) is:

a. –mv
b. −Mv/m
c. −mv/M
d. –v

MCQ # 12. A 4.0-N puck is traveling at 3.0m/s. It strikes a 8.0-N puck, which is stationary. The two pucks stick together. Their common final speed is:

a. 1.0m/s
b. 1.5m/s
c. 2.0m/s
d. 2.3m/s

MCQ # 13. One revolution is the same as:

a. 1 rad
b. 57 rad
c. π/2 r
d. 2π rad

MCQ # 14. If force is applied at center of mass then torque is:
a. Infinity
b. 1
c. Zero
d. Maximum

MCQ # 15. The fundamental dimensions of angular momentum are
a. $ML^{-2}T^{-2}$
b. MLT^{-1}
c. $ML2T^{-1}$
d. $ML2T^{-2}$

MCQ # 16. A 2.0-kg block travels around a 0.50-m radius circle with an angular velocity of 12 rad/s. Its angular momentum about the center of the circle is:
a. 6.0 kg x m^2/s
b. 12 kg x m^2/s
c. 8 kg/m^2 x s
d. 72 kg x m^2/s^2

MCQ # 17. The center of gravity is the average location of the …….. of an object
a. Mass
b. Weight
c. static equilibrium
d. dynamic

MCQ # 18. In simple harmonic motion, the restoring force must be proportional to the:
a. Amplitude
b. Frequency
c. Velocity
d. Displacement

MCQ # 19. Take the speed of sound to be 340m/s. A thunder clap is heard about 3 s after the lightning is seen. The source of both light and sound is:
a. moving overhead faster than the speed of sound
b. emitting a much higher frequency than is heard
c. emitting a much lower frequency than is heard
d. about 1000m away

MCQ # 20. Young's modulus can be correctly given in:
a. Nm
b. N/m^2
c. N/m
d. Nm/s

MCQ # 21. A bucket of water is pushed from left to right with increasing speed across a horizontal surface. Consider the pressure at two points at the same level in the water.
a. It is the same
b. It is higher at the point on the left

c. It is higher at the point on the right

d. At first it is higher at the point on the left but as the bucket speeds up it is lower there

MCQ # 22. An organ pipe with both ends open is 0.85m long. Assuming that the speed of sound is 340m/s, the frequency of the third harmonic of this pipe is

a. 200 Hz

b. 300 Hz

c. 400 Hz

d. 600 Hz

MCQ # 23. Does the pitch get higher, lower or does it stay the same as the amplitude increases?

a. Higher

b. Lower

c. Stays the same

d. None of these

MCQ # 24. An astronaut on the Moon simultaneously drops a feather and a hammer. The fact that they land together shows that:

a. no gravity forces act on a body in a vacuum

b. the acceleration due to gravity on the Moon is less than on Earth

c. in the absence of air resistance all bodies at a given location fall with the same acceleration

d. the feather has a greater weight on the Moon than on Earth

MCQ # 25. Typical value for the magnitude of the electric field inside the atom is

a. 10-11N/C

b. 1011N/C

c. 10-9N/C

d. 109N/C

MCQ # 26. If the distance between two opposite charges is increased by a factor of 3, the Coulomb s' force of attraction between them will

a. reduce by a factor of 3

b. reduce by a factor of 9

c. remain Same

d. increase by a factor of 3

Attempt any 4 question of the following 3 marks questions

Desc 1. Is it possible that action and reaction are always equal? If yes, why does the body accelerate?

Desc 2. Discuss Kepler' 3rd law of planetary motion, also write its mathematical formula.

Desc 3. When do you usually drop a ball from a certain height it accelerates downward at 9.8 m/s^2? If you instead throw it downward, then its acceleration instantly after departure your hand, assuming no air resistance, is?

Desc 4. In which type of frame of reference Newton s laws holds. Give your opinion about the Earth, Either it is inertial or non-inertial.

Desc 5. The whole electrical technology revolves round the devices i.e. motor, generator, and transformer. Write your answer in a single line that under what relationship these devices came into existence.

Desc 6. Presume a ship cruises with a constant velocity when the thrust from its engines is a constant 10000N. What is the acceleration of the ship? What is the force of air resistance acting on the ship?

Attempt any 3 question of the following 5 marks questions

Desc 1. Calculate the rotational kinetic energy of the Sphere.

Desc 2. Calculate the magnitude of the angular momentum of the earth considered as a particle orbiting the sun. The mass of the earth is 5.97×10^{24} kg. Treat it as moving in a circular orbit of radius 1.50×10^{11} m at a speed of 2.98×10^{4} m/s.

Desc 3. A juggler (as shown above) throws each ball so it hangs in the air for 1.60seconds before landing in the other hand, 75cm away. What are the initial vertical and horizontal velocity components?

Desc 4. The coefficient of kinetic friction is 0.300. What is the magnitude of the tension force in the rope (shown in the figure below)?

